

# Cell3 to Mass-Bridge

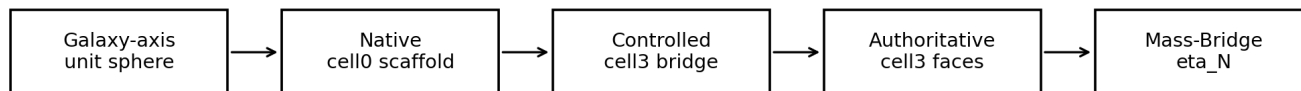
Educational Derivation from Galaxy-Sourced cell0 and the Authoritative cell3 Bridge

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| Field                 | Value                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Generated file        | cell3_to_mass_bridge_1783390310.pdf                                                                                                |
| Unix epoch            | 1783390310                                                                                                                         |
| Derivation route      | Galaxy-axis unit sphere -> native galaxy cell0 -> controlled perturbation to authoritative cell3 -> primitive faces -> Mass-Bridge |
| Educational objective | Derive the Mass-Bridge from the authoritative cell3 geometry while preserving the galaxy-sourced cell0 provenance.                 |

This paper derives the Mass-Bridge as a face-area relation of authoritative cell3. The native galaxy-sourced cell0 establishes provenance from galaxy-axis data. The cell0-to-cell3 bridge then supplies the controlled geometric transition needed for the closure geometry used in all downstream formulas. The Mass-Bridge itself is computed only after the authoritative cell3 face areas are fixed.



Provenance chain used in this derivation

Figure 1. Provenance chain used in the Mass-Bridge derivation.

## Abstract

The Mass-Bridge is the scaled ratio of the authoritative cell3 primitive face area  $A_{bc}$  to the primitive face area  $A_{ab}$ . The derivation proceeds through a documented provenance chain: galaxy-axis group centroids are converted into a native unit-sphere population, the mirror-accumulator method recovers a native galaxy cell0 scaffold, and a controlled perturbation carries that scaffold to authoritative cell3. From authoritative cell3, the primitive face areas are  $A_{ab} = 7.547217896398894727e-04$ ,  $A_{ac} = 1.011255757656946046e-03$ , and  $A_{bc} = 1.385784431740137712e-03$ . The Mass-Bridge is  $\eta_N = 1000 \cdot A_{bc}/A_{ab} = 1836.152673425999183$ .

The same face triangle gives the companion channels  $R_{acab} = A_{ac}/A_{ab} = 1.339905341992922$ ,  $C_{\alpha} = A_{bc}/A_{ac} = 1.370359991770000$ ,  $\alpha_N = A_{ac}/(100 \cdot A_{bc}) = 0.007297352564331425$ , and  $c_N = 1/\alpha_N = 137.035999177000008$ . These are not independent assignments. They are reciprocal readings of the same three primitive faces.

| Section                                     | Purpose                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------|
| 1. Galaxy-sourced cell0 provenance          | Show the native galaxy-derived scaffold and the coordinate convention used to construct it. |
| 2. Controlled bridge to authoritative cell3 | Show the small geometric transition from cell0 to cell3.                                    |
| 3. Authoritative cell3 face layer           | Define and display the three primitive face areas.                                          |
| 4. Mass-Bridge derivation                   | Compute $\eta_N$ from $A_{bc}/A_{ab}$ .                                                     |
| 5. Companion face-channel closure           | Compute the carrier-triangle identities and checks.                                         |
| 6. Final Mass-Bridge figure                 | End with a dimensionally correct geometric figure of the Mass-Bridge.                       |

## 1. Galaxy-sourced cell0 provenance

The galaxy-axis source is first expressed as a unit-sphere population. The native cell recovered by the mirror-accumulator method is cell0, the original galaxy-derived rhombohedral scaffold. At this stage, authoritative cell3 is not an input; cell0 is the result of the galaxy-sourced recovery process.

| Object                   | Value                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------|
| source data rows         | 126006                                                                                           |
| unique group centroids   | 1000                                                                                             |
| Dec range                | 0.088361 to 87.522719 deg                                                                        |
| z range before antipodes | 0.001542190338 to 0.999065438844                                                                 |
| unit-vector conversion   | $x=\cos(\text{dec})\cos(\text{ra})$ , $y=\cos(\text{dec})\sin(\text{ra})$ , $z=\sin(\text{dec})$ |
| antipodal construction   | $X = \{x_i, -x_i\}$ , performed after source conversion                                          |

| Quantity     | cell0 value              |
|--------------|--------------------------|
| a            | 0.020788448186           |
| b            | 0.036478622795           |
| c            | 0.049350062300           |
| theta_ab     | 83.928361201001 deg      |
| theta_ac     | 80.452247903308 deg      |
| theta_bc     | 50.382886569535 deg      |
| $A_{ab}$     | 7.540800268679999753e-04 |
| $A_{ac}$     | 1.01169998599999982e-03  |
| $A_{bc}$     | 1.386752323000000004e-03 |
| volume       | 2.842908885472000073e-05 |
| surface area | 0.006305064672           |

## 2. Controlled bridge to authoritative cell3

The native cell0 geometry is a galaxy-sourced scaffold. The Mass-Bridge used in the closure chain is computed after the controlled transition to authoritative cell3. The bridge is a small geometric perturbation of lengths, angles, face areas, volume, and surface area. This preserves provenance while placing the final computation on the closure geometry.

| Quantity | cell0              | cell3              | delta cell3-cell0   | relative delta      |
|----------|--------------------|--------------------|---------------------|---------------------|
| a        | 0.020788448186     | 0.020797058782     | 8.610595999999e-06  | 4.142009986968e-04  |
| b        | 0.036478622795     | 0.036494205131     | 1.558233600000e-05  | 4.271634948383e-04  |
| c        | 0.049350062300     | 0.049308565900     | -4.149640000000e-05 | -8.408581076908e-04 |
| theta_ab | 83.928361201001    | 83.933483012313    | 0.005121811312      | 6.102598976933e-05  |
| theta_ac | 80.452247903308    | 80.448130018338    | -0.004117884970     | -5.118421271397e-05 |
| theta_bc | 50.382886569535    | 50.363231813756    | -0.019654755779     | -3.901077750255e-04 |
| A_ab     | 7.540800268680e-04 | 7.547217896399e-04 | 6.417627719000e-07  | 8.510539319885e-04  |
| A_ac     | 1.011699986000e-03 | 1.011255758000e-03 | -4.442280000000e-07 | -4.390906455938e-04 |
| A_bc     | 1.386752323000e-03 | 1.385784432000e-03 | -9.678910000001e-07 | -6.979552036417e-04 |

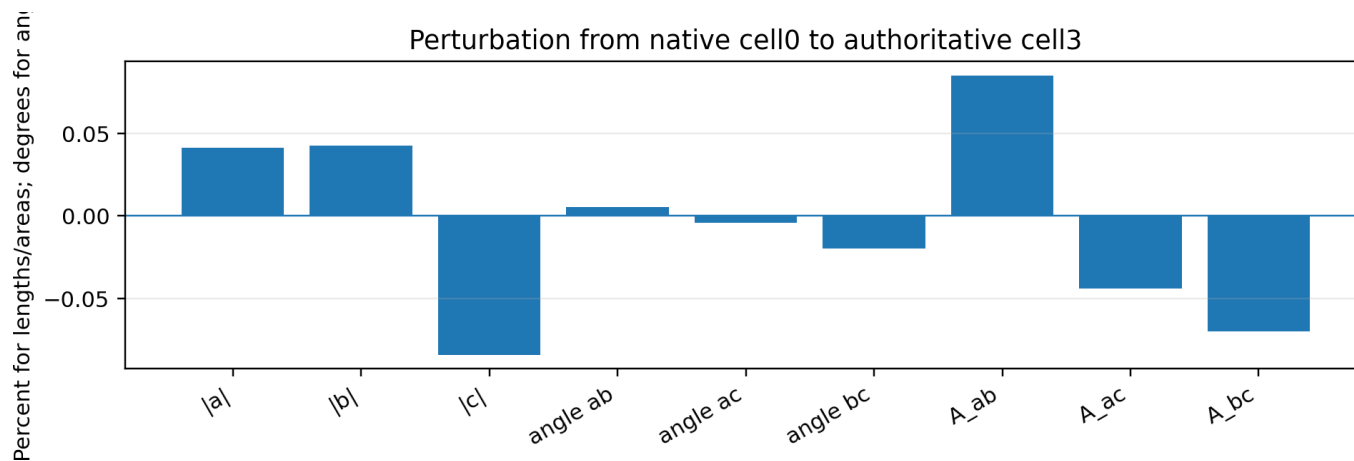


Figure 2. Measured transition from native cell0 to authoritative cell3.

## 3. Authoritative cell3 primitive face layer

Let a, b, and c be the three primitive edge vectors of authoritative cell3. Each pair of edges defines a primitive parallelogram face. The three face areas are computed by cross products, or equivalently by the parallelogram rule using side lengths and included angles.

$$\begin{aligned} A_{ab} &= |a \times b| \\ A_{ac} &= |a \times c| \\ A_{bc} &= |b \times c| \end{aligned}$$

| Face | Side lengths and included angle                                     | Area                     |
|------|---------------------------------------------------------------------|--------------------------|
| ab   | a =0.020797058782,  b =0.036494205131, theta_ab=83.933483012313 deg | 7.547217896398894727e-04 |
| ac   | a =0.020797058782,  c =0.049308565900, theta_ac=80.448130018338 deg | 1.011255757656946046e-03 |
| bc   | b =0.036494205131,  c =0.049308565900, theta_bc=50.363231813756 deg | 1.385784431740137712e-03 |

Authoritative cell3 primitive faces at common scale

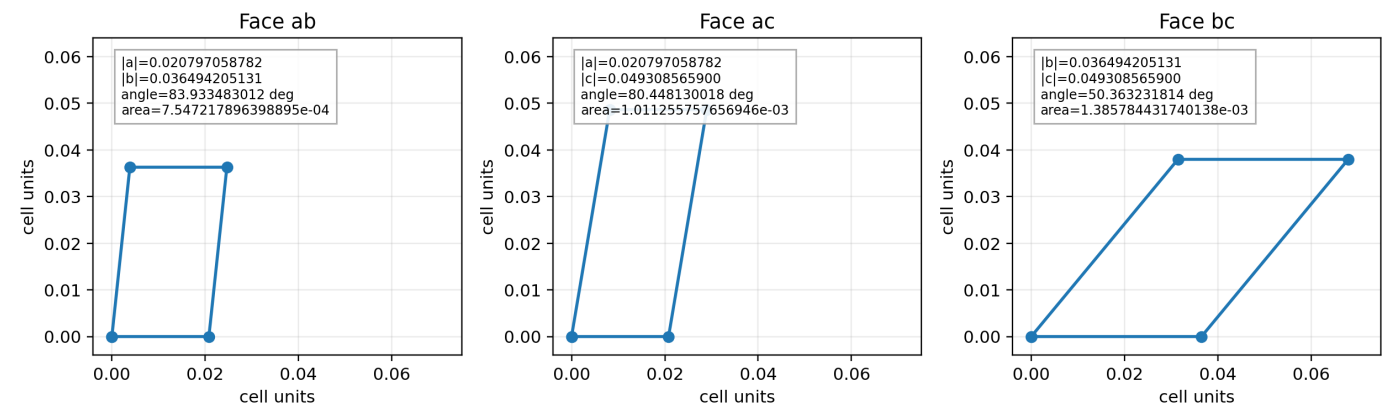


Figure 3. Authoritative cell3 primitive faces drawn at common scale.

## 4. Mass-Bridge derivation

The Mass-Bridge is the scaled ratio of the bc primitive face channel to the ab primitive face channel. The denominator face is ab. The numerator face is bc. The scale factor 1000 is the native convention used for the mass-channel reading.

$$C_m = A_{bc} / A_{ab}$$

$$\text{eta}_N = 1000 * C_m = 1000 * A_{bc}/A_{ab}$$

Substitution gives:

$$C_m = 1.385784431740137712e-03 / 7.547217896398894727e-04 = 1.836152673425999193$$

$$\text{eta}_N = 1000 * 1.836152673425999193 = 1836.152673425999183$$

The authoritative cell3 Mass-Bridge is  $\text{eta}_N = 1836.152673425999183$ .

| Step | Quantity          | Formula                 | Value                    |
|------|-------------------|-------------------------|--------------------------|
| 1    | Denominator face  | $A_{ab}= a \times b $   | 7.547217896398894727e-04 |
| 2    | Numerator face    | $A_{bc}= b \times c $   | 1.385784431740137712e-03 |
| 3    | Raw mass quotient | $C_m=A_{bc}/A_{ab}$     | 1.836152673425999193     |
| 4    | Mass-Bridge       | $\text{eta}_N=1000*C_m$ | 1836.152673425999183     |

## 5. Companion face-channel closure

The same three faces also produce a closed carrier triangle. These companion values preserve the connection between the mass channel, cross-leg channel, and fine/speed channel.

$$R_{acab} = A_{ac}/A_{ab}$$

$$C_{\alpha} = A_{bc}/A_{ac}$$

$$R_{acab}*C_{\alpha} = A_{bc}/A_{ab} = C_m$$

$$\alpha_N = A_{ac}/(100*A_{bc})$$

$$c_N = 1/\alpha_N$$

| Quantity              | Definition                  | Value                |
|-----------------------|-----------------------------|----------------------|
| $R_{acab}$            | $A_{ac}/A_{ab}$             | 1.339905341992921661 |
| $C_{\alpha}$          | $A_{bc}/A_{ac}$             | 1.370359991770000008 |
| $R_{acab}*C_{\alpha}$ | Closure product             | 1.836152673425999193 |
| $C_m$                 | $A_{bc}/A_{ab}$             | 1.836152673425999193 |
| Closure residual      | $R_{acab}*C_{\alpha} - C_m$ | 1.000e-49            |
| $\alpha_N$            | $A_{ac}/(100*A_{bc})$       | 0.007297352564331425 |
| $c_N$                 | $1/\alpha_N$                | 137.0359991770000008 |

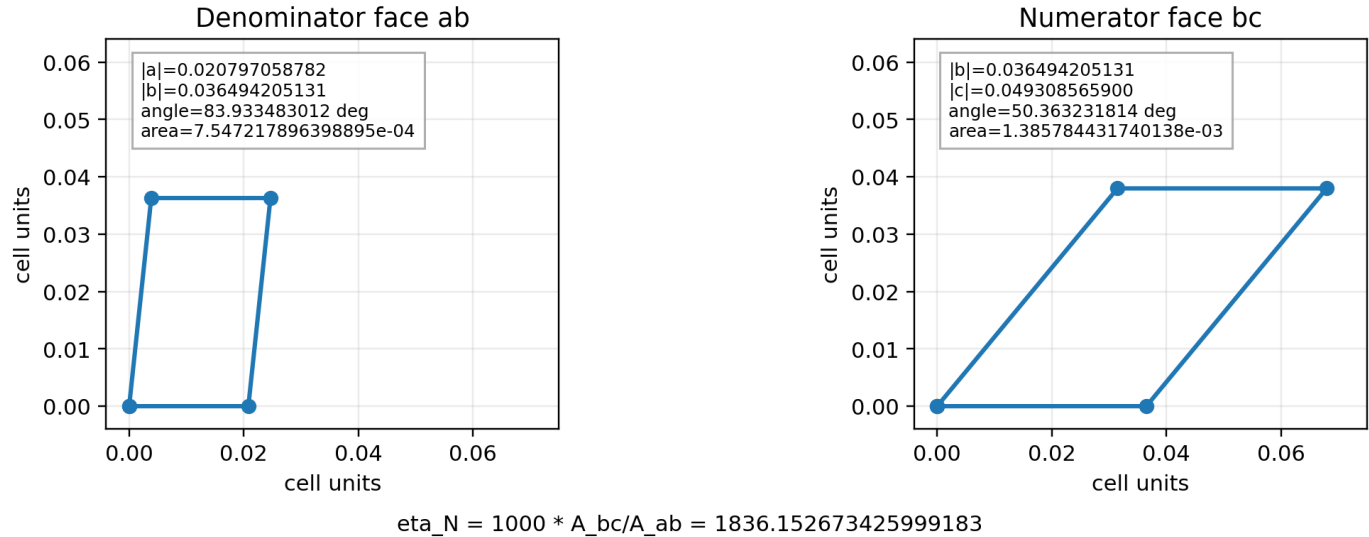
These relationships show that the Mass-Bridge is one reading of the same primitive face system that also supplies the fine/speed channel.

## 6. Final Mass-Bridge figure

The final figure places the denominator and numerator faces side by side at common cell-unit scale. The face geometries are drawn from the authoritative cell3 side lengths and included angles, so the figure is dimensionally correct in native cell units.

$$\text{eta}_N = 1000 * (A_{bc}/A_{ab}) = 1000 * (1.385784431740138e-03/7.547217896398895e-04) = 1836.152673425999183$$

## Dimensionally correct Mass-Bridge construction



Final Figure. Dimensionally correct Mass-Bridge construction from authoritative cell3 faces ab and bc.

Final reconstruction statement: galaxy-axis data produce native cell0; the controlled bridge produces authoritative cell3; authoritative cell3 gives  $A_{ab}$ ,  $A_{ac}$ , and  $A_{bc}$ ; and the scaled face quotient gives  $\text{eta}_N = 1836.152673425999183$ .